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A Modified Flux Observer for Sensorless Direct Torque Control of Dual Three-Phase PMSM Considering Open-Circuit Fault

Xianqun Qiu, Jinghua Ji, Dao Zhou, *Senior Member, IEEE*, Wenxiang Zhao, *Senior Member, IEEE*, Yifan Chen, and Linsen Huang

Abstract—This paper proposes a modified flux observer for sensorless direct torque control (DTC) of the dual three-phase permanent magnet synchronous motor (PMSM). The proposed method can improve torque performance without position sensor. The flux observer is proposed to obtain the stator fluxes and rotor fluxes. The stator fluxes can be utilized to calculate the amplitude of torque and flux for DTC system. Meanwhile, the position information can be detected from the rotor fluxes by using a normalized phase-locked loop. This new flux observer can overcome the disadvantages due to the dc bias, initial value, and harmonic components. The stability of the proposed observer is also analyzed in this paper. In addition, a fault-tolerant DTC scheme for open-circuit fault is introduced. The performance of torque control and position estimation are evaluated under healthy, open-circuit fault, and fault-tolerant conditions. Simulation and experimental evaluation are carried out to confirm the effectiveness of the proposed sensorless DTC scheme.

Index Terms—Flux observer, direct torque control (DTC), sensorless control, fault-tolerant, permanent-magnet synchronous motor (PMSM), dual three-phase motor.

I. INTRODUCTION

MULTI-PHASE motor drives have attracted significant attention due to the development of control theory and power electronics. Compared with three-phase motor drives, multi-phase motor drives have the merits of high-power, low dc-link voltage requirement, high reliability, and excellent fault-tolerant capability [1]. As the most common type among different types of multi-phase motor, the dual three-phase permanent-magnet synchronous motor (PMSM) was discussed in some literatures [2]–[4]. Usually, the dual three-phase PMSM has two stator windings sets spatially shifted by $\pi/6$ with isolated neutral points, and it has the intrinsic feature of elimination of sixth harmonic torque pulsation [5].

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Direct torque control (DTC) has the advantages of rapid torque response, simple structure, and less parameter dependence [6]–[8]. However, the conventional switching-table DTC suffers from some shortcomings such as variable switching frequency and large torque ripple. Instead, the space vector modulation based DTC was proposed due to the advantages of accurate control for torque and flux, reliable low-speed and fixed switching frequency [9]. In the space vector modulation based DTC system, the stator flux is an important parameter which seriously affects the control performance. Meanwhile, the accurate rotor position is needed for both the calculation of stator flux based on current model and motor speed for precise torque control [10][11]. The mechanical position sensor is commonly used to obtain the rotor position and speed. However, the adoption of mechanical sensor not only increases the cost and size, but also reduces the reliability and robustness of drive system. To overcome these drawbacks, sensorless control has attracted much attention [12]–[18]. In general, the model-based method is common in sensorless control. The model-based method for sensorless control is based on the estimation of back electromotive force (EMF) or rotor flux, and it is suitable for medium-high speed operation [19].

Usually, the stator flux can be directly calculated by using the voltage model or the current model [20]. However, it is inevitable to encounter several problems such as parameter uncertainties, dc bias, the inverter nonlinearity effect, and noise [21]. In [22], an improved nonlinear discrete stator flux estimator was developed to replace the pure integrator. The flux estimator can reduce the initial value and dc bias problems. A wide speed sliding mode observer was adopted in [23], which can be used to estimate the back EMF for sensorless DTC system. However, the system chattering problem is introduced. In [24], a model reference adaptive system estimator based on the second-order sliding-mode strategy was introduced, which can overcome the chattering problem, parameter uncertainties, and dc bias. A flux observer based on the Gopinath model was proposed in [25], which can eliminate the effects of dc bias and initial value. However, the flux estimation result will fluctuate at the boundary frequency between the current model and the voltage model.

In addition, model-based sensorless methods are utilized to achieve more accurate position information when motor operates on medium-high speed region. In [26], a fifth order

flux observer for sensorless drive was proposed. The rotor flux observer is designed to suppress dc bias and high order harmonics, then improve the position estimation accuracy. In [27], a flux linkage observer for sensorless direct thrust force control was presented, which has strong robustness against the interferences, no phase lag, and no attenuation. Meanwhile, an extra back EMF observer is designed to obtain the rotor position. A novel sensorless flux adaption direct torque and flux control for interior PMSM was investigated in [28]. In order to achieve maximum torque per ampere and direct torque and flux control synchronously, the amplitude of stator flux is directly calculated based on the existing active flux. By using a second-order sliding mode observer, the back EMF can be obtained to estimate the rotor position. Since the amplitude of rotor flux is not related to velocity, the sensorless method based on rotor flux is superior to the method based on back EMF [29]. However, only the healthy condition is considered in these methods. Among all types of faults, the open-circuited failures of motor phases are the most common scenarios [30][31]. Meanwhile, the disturbances is unexpected for motor control when the open-phase fault occurs. Due to the loss of one phase current, the current distortion and torque fluctuation of motor will become more serious which brings challenge for torque control and position estimation. Furthermore, some fault-tolerant DTC schemes based on reduced decoupling matrix is relatively complex in the switching process, which make it difficult for sensorless drives [32].

In this paper, a modified flux observer for sensorless DTC of the dual three-phase PMSM will be proposed. The healthy, open-circuit fault, and fault-tolerant conditions are considered when evaluating the performance of torque control and position estimation. The proposed flux observer is designed to obtain the stator fluxes and rotor fluxes, which can overcome the demerits of current and voltage dc bias, initial value, and the high frequency noise. The estimated stator fluxes can be applied to the DTC scheme for better torque performance. Meanwhile, the estimated rotor fluxes contains position information which can be used for sensorless operation. A sensorless DTC system with the proposed flux observer is introduced and verified by the simulation and experiment. The rest of this paper is organized as follows. Section II briefly describes the mathematical model of the dual three-phase PMSM. The proposed flux observer is presented in Section III. Based on the proposed flux observer, the sensorless DTC system is presented in Section IV. Section V provides simulation and experimental results to evaluate the performance of the proposed flux observer. Finally, the conclusions are presented in Section VI.

II. MATHEMATICAL MODEL

According to the modelling technique, the dual three-phase PMSM is modelled in three subspaces, which are commonly named as α - β , z_1 - z_2 , and o_1 - o_2 subspaces. However, only the α - β components share useful electromechanical energy conversion. The z_1 - z_2 , and o_1 - o_2 components cannot generate any usable electromechanical energy and just produce losses. The transformation between the normal dual three-phase system

(i.e., abc - def) and the decoupled subspaces (i.e., α - β , z_1 - z_2 , and o_1 - o_2) is performed using the matrix as

$$T_{6s} = \frac{1}{3} \begin{bmatrix} 1 & -\frac{1}{2} & -\frac{1}{2} & \frac{\sqrt{3}}{2} & -\frac{\sqrt{3}}{2} & 0 \\ 0 & \frac{\sqrt{3}}{2} & -\frac{\sqrt{3}}{2} & \frac{1}{2} & \frac{1}{2} & -1 \\ 1 & -\frac{1}{2} & -\frac{1}{2} & -\frac{\sqrt{3}}{2} & \frac{\sqrt{3}}{2} & 0 \\ 0 & -\frac{\sqrt{3}}{2} & \frac{\sqrt{3}}{2} & \frac{1}{2} & \frac{1}{2} & -1 \\ 1 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 & 1 \end{bmatrix} \quad (1)$$

where the fundamental components and the harmonics of the order $12k \pm 1$, ($k = 1, 2, 3, \dots$) in the dual three-phase PMSM are mapped into the α - β subspace, which corresponding with the first and second rows of the matrix. The harmonics of the order $6k \pm 1$, ($k = 1, 3, 5, \dots$) are transferred into the z_1 - z_2 subspace, and the zero sequence harmonics are mapped into the o_1 - o_2 subspace. The z_1 - z_2 subspace and o_1 - o_2 subspace are corresponding to the middle two rows and the last two rows of the matrix, respectively.

Assuming sinusoidal winding distribution and neglecting core losses and magnetic saturation, the final model of the dual three-phase PMSM can be built. The voltage equations of the dual three-phase PMSM in the α - β subspace can be expressed as

$$\begin{cases} U_\alpha = R_s i_\alpha + \frac{d\psi_{s\alpha}}{dt} \\ U_\beta = R_s i_\beta + \frac{d\psi_{s\beta}}{dt} \end{cases} \quad (2)$$

where $\psi_{s\alpha}$ and $\psi_{s\beta}$ are the stator fluxes of $\alpha\beta$ -axis, i_α and i_β are the currents of $\alpha\beta$ -axis, R_s is the stator resistance of motor. The flux equations are given by

$$\begin{cases} \psi_{s\alpha} = \psi_{f\alpha} + L_s i_\alpha \\ \psi_{s\beta} = \psi_{f\beta} + L_s i_\beta \end{cases} \quad (3)$$

with

$$\begin{cases} |\psi_s| = \sqrt{\psi_{s\alpha}^2 + \psi_{s\beta}^2} \\ \theta_s = \arctan \frac{\psi_{s\beta}}{\psi_{s\alpha}} \end{cases} \quad (4)$$

where $\psi_{f\alpha}$ and $\psi_{f\beta}$ are the rotor fluxes of $\alpha\beta$ -axis, L_s is the stator inductance of motor, $|\psi_s|$ is the amplitude of stator flux, and θ_s is the stator flux angle.

The rotor fluxes can be regarded as the distribution of the permanent magnet flux linkage based on the rotor position. Then, it can be derived as

$$\begin{cases} \psi_{f\alpha} = \psi_{pm} \cos \theta_e \\ \psi_{f\beta} = \psi_{pm} \sin \theta_e \end{cases} \quad (5)$$

$$\theta_e = \arctan \frac{\psi_{f\beta}}{\psi_{f\alpha}} \quad (6)$$

where ψ_{pm} is the permanent magnet flux linkage, and θ_e is the electrical angular position.

The voltage equations of the dual three-phase PMSM in the z_1 - z_2 subspace and o_1 - o_2 subspace are presented as

$$U_{z_1 z_2} = R_s i_{z_1 z_2} + \frac{d\psi_{z_1 z_2}}{dt} \quad (7)$$

$$U_{o_1 o_2} = R_s i_{o_1 o_2} + \frac{d\psi_{o_1 o_2}}{dt} \quad (8)$$

where $i_{z_1 z_2}$ represents the currents in the z_1 - z_2 subspace, and $i_{o_1 o_2}$ represents the currents in the o_1 - o_2 subspace. $\psi_{z_1 z_2}$ and $\psi_{o_1 o_2}$ are the flux linkages in the z_1 - z_2 subspace and o_1 - o_2 subspace, respectively. In order to minimize the losses caused by the harmonic currents, the currents in the z_1 - z_2 subspace also needs to be controlled separately. Since there is no circulating current in the two sets of windings with isolated neutral points, the currents in the z_1 - z_2 subspace can be ignored.

The electromagnetic torque of the dual three-phase PMSM is defined as

$$T_e = 3p_n (\psi_{sa} i_\beta - \psi_{s\beta} i_a) \quad (9)$$

where p_n is the number of the pole pairs.

III. PROPOSED FLUX OBSERVER

In the conventional method, the flux is calculated directly by the voltage model or the current model. However, it will inevitably brings the issues of accuracy and reliability. When the currents and voltages are measured by sensors, the dc bias and high-frequency noise are also unavoidable. Furthermore, the effect of dc bias will be further amplified with a pure integrator. Therefore, the aim of this section is to design a novel flux observer for accurate estimation of stator flux and rotor flux which can be used to build the sensorless DTC system.

According to (2) and (3), the stator flux and rotor flux equations can be modified as

$$\begin{cases} \psi_{sa} = \frac{1}{s} (U_a - R_s i_a) \\ \psi_{s\beta} = \frac{1}{s} (U_\beta - R_s i_\beta) \end{cases} \quad (10)$$

$$\begin{cases} \psi_{fa} = \frac{1}{s} (U_a - R_s i_a) - L_s i_a \\ \psi_{f\beta} = \frac{1}{s} (U_\beta - R_s i_\beta) - L_s i_\beta \end{cases} \quad (11)$$

The structure of the proposed flux observer is shown in Fig. 1. It can be seen that the voltage $u_{a\beta}$ and the current $i_{a\beta}$ are the inputs, and stator flux $\psi_{sa\beta}$ and rotor flux $\psi_{fa\beta}$ are the outputs. G_3 is the feedback controller. G_1 and G_2 are the disturbance observer which can be expressed as

$$G_1(s) = G_2(s) = 1 - D(s) \quad (12)$$

with

$$D(s) = \frac{s^2 + \omega_0^2}{s^2 + \xi\omega_0 s + \omega_0^2} \quad (13)$$

where $D(s)$ can be regarded as a notch filter and its output is the component of high frequency noise and dc bias. ω_0 is the electrical angular velocity which is consistent with the frequency of flux. The parameter ξ is the bandwidth gain of the

notch filter. Thus, the outputs of G_1 and G_2 are the signal of flux without high frequency noise and dc bias.

The proposed flux observer can be regarded as a combination of a stator flux observer and a rotor flux observer. Thus, the transfer function of stator flux observer can be expressed as

$$\begin{aligned} H_s(s) &= \frac{\psi_{sa\beta}}{\frac{1}{s}(u_{a\beta} - R_s i_{a\beta})} \\ &= \frac{L_s(1-D(s))}{L_s + G_3[(1-D(s))^2 - 1]} \end{aligned} \quad (14)$$

then, the transfer function of rotor flux observer can be expressed as

$$\begin{aligned} H_f(s) &= \frac{\psi_{fa\beta}}{\frac{1}{s}(u_{a\beta} - R_s i_{a\beta}) - L_s i_{a\beta}} \\ &= \frac{L_s(1-D(s))^2}{L_s + G_3[(1-D(s))^2 - 1]} \end{aligned} \quad (15)$$

meanwhile, the convergence rate is decided by feedback controller which can be defined as

$$G_3(s) = g \quad (16)$$

where g is the feedback gain. The feedback gain can not only accelerate the convergence process, but also adjust the bandwidth of the flux observer. The error between estimated current and real current of $\alpha\beta$ -axis can be amplified by the feedback gain. When $g = 0$, the observer can be equivalent to an open-loop system. Due to the existence of the pure integrator, the dc bias will lead to the failure of observer in the open-loop system. Therefore, the feedback gain should be set appropriately to ensure the stability of the system.

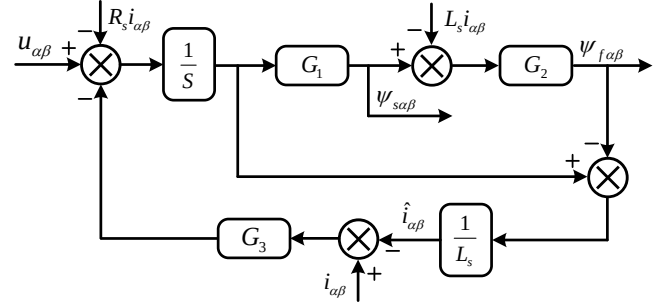


Fig. 1. Structure diagram of proposed flux observer.

In order to evaluate the performance of stator flux and rotor flux observer, the stability of observers should be investigated. According to (14) and (15), the transfer function of stator flux observer and rotor flux observer can be modified as

$$H_s(s) = \frac{(s^2 + \xi\omega s + \omega^2)L_s\xi\omega s}{(L_s - g)s^4 + 2\xi\omega(L_s - g)s^3 + \omega^2(2L_s - 2g + \xi^2L_s)s^2 + 2\xi\omega^3(L_s - g)s + (L_s - g)\omega^4} \quad (17)$$

$$H_f(s) = \frac{L_s\xi^2\omega^2s^2}{(L_s - g)s^4 + 2\xi\omega(L_s - g)s^3 + \omega^2(2L_s - 2g + \xi^2L_s)s^2 + 2\xi\omega^3(L_s - g)s + (L_s - g)\omega^4} \quad (18)$$

Based on (17) and (18), it can be seen that the characteristic equation of two flux observers are the same. All the parameters of characteristic equation should be positive to ensure the stability of the system. The characteristic equation can be expressed as

$$d(s) = (L_s - g)s^4 + 2\xi\omega(L_s - g)s^3 + \omega^2(2L_s - 2g + \xi^2 L_s)s^2 + 2\xi\omega^3(L_s - g)s + (L_s - g)\omega^4 \quad (19)$$

meanwhile, the Routh array can be expressed as

$$\begin{array}{c|cc} s^4 & L_s - g & 2(L_s - g) + \xi^2 L_s & (L_s - g)\omega^4 \\ s^3 & 2(L_s - g)\xi\omega & 2(L_s - g)\xi\omega^3 & \\ s^2 & (L_s - g + \xi^2 L_s)\omega^2 & (L_s - g)\omega^4 & \\ s^1 & \frac{2(L_s - g)L_s\xi^3\omega}{L_s - g + \xi^2 L_s} & & \\ s^0 & (L_s - g)\omega^4 & & \end{array} \quad (20)$$

In terms of the Routh stability criterion, the stability of observer can be guaranteed when the elements of first column of the Routh array are all positive. Thus, the feedback gain should be less than the stator inductance.

The disturbance rejection and noise suppression are analyzed by the nyquist plot and bode diagram, as shown in Figs. 2 and 3. The center frequency of observer is set to 20 Hz, the bandwidth is set to 4 Hz, and the feedback gain is set to -0.92. As shown in Fig. 2, the stator flux observer and rotor flux observer have the similar trajectories, which do not include the point (-1, j0). Thus, the stability of stator flux observer and rotor flux observer can be guaranteed. In Fig. 3, the stator flux observer and rotor flux observer are similar to the bandpass filter. The bandpass filter can effectively suppress other components in the signal except the center frequency. It can be seen that the amplitude decreases with the distance from the center frequency. The smaller feedback gain reduces the bandwidth of the stator flux observer and rotor flux observer at the same time. Therefore, both the stator flux and rotor flux observer have the ability to reduce disturbance caused by the noise, initial value, and dc bias.

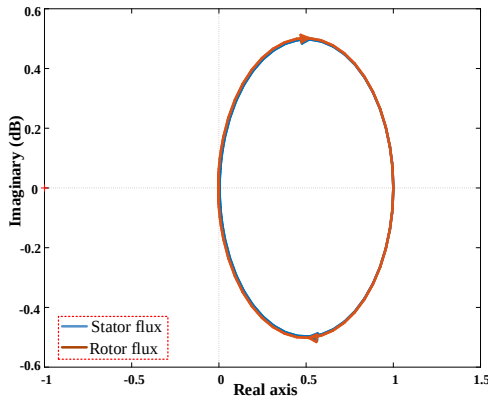


Fig. 2. Nyquist plot of proposed observer for stator flux and rotor flux.

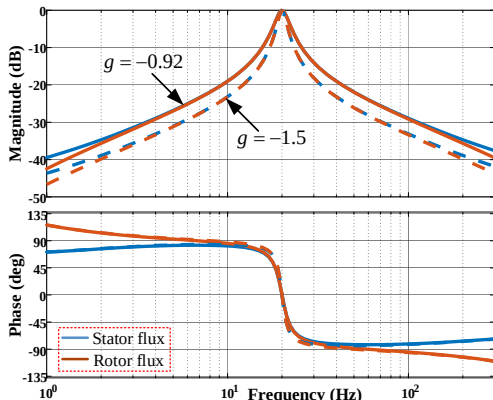


Fig. 3. Bode diagram of proposed observer for stator flux and rotor flux.

IV. SENSORLESS DTC SYSTEM

By utilizing the proposed flux observer, the stator fluxes and rotor fluxes can be obtained. Then, the amplitude of stator flux and torque can be calculated, which will join the DTC scheme eventually. Meanwhile, the rotor flux contains the position information which can be extracted by the position observer. A fault-tolerant DTC scheme for the open-circuit fault condition and position observer based on the phase-locked loop are presented in this section.

A. Fault-Tolerant DTC Scheme

Fig. 4 shows the topology diagram of dual three-phase motor under open-circuit fault condition. In this paper, the connection between the phase F winding and the voltage source inverter is disconnected when an open-circuit fault occurs. Thus, the current of phase F becomes zero ($i_f = 0$), and the machine loses one degree of freedom. Since the motor's phase winding is not damaged during fault, only the current will be affected if the decoupling transformation matrix remains unchanged.

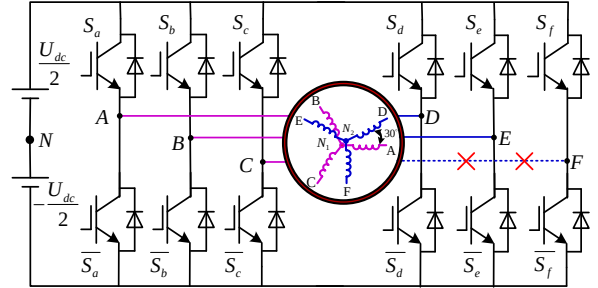


Fig. 4. Topology diagram of dual three-phase motor under one-phase fault.

By adjusting the reference current of healthy phases, the maximum torque operation of motor under open-circuit fault condition can be realized. In order to keep the motor magnetomotive force unchanged, the reference current of healthy phases can be expressed as

$$\begin{cases} i_a = 0 \\ i_b = \sqrt{3}I_m \cos(\theta_e - 90^\circ) \\ i_c = \sqrt{3}I_m \cos(\theta_e + 90^\circ) \\ i_d = \sqrt{3}I_m \cos(\theta_e) \\ i_e = \sqrt{3}I_m \cos(\theta_e - 180^\circ) \end{cases} \quad (21)$$

where I_m is the current amplitude of dual three-phase motor during normal operation, i_a , i_b , i_c , i_d , and i_e represents the current of phase A, phase B, phase C, phase D, and phase E, respectively.

However, the application range of fault-tolerant control strategy based on this hysteresis current control is limited. Based on the normal decoupling transformation, an open-circuit fault-tolerant control method is introduced. According to (1), only the currents of i_β , i_{z2} , and i_{o2} can be affected by the current of phase F. The relationship can be expressed as

$$i_f = -i_\beta + i_{z2} + i_{o2} = 0 \quad (22)$$

if the motor operates with an open-circuit fault, and the transformation matrix under normal state is still used for calculation. It is clear that the currents in the fundamental and harmonic subspace are no longer decoupled. Hence, the torque ripple is inevitable if the reference current of harmonic

subspace is set to zero. As shown in Fig. 4, the dual three-phase motor has two isolated neutral points. Since the currents in the α_1 - α_2 subspace cannot flow, the zero-sequence current controllers can be disabled by setting $i_{\alpha 1} = i_{\alpha 2} = 0$. The torque equation is not affected by the fault, as indicated in (9). In order to achieve smooth torque during fault, the reference currents in the α - β subspace should be set as

$$\begin{cases} i_{\alpha}^* = I_m \cos(\theta) \\ i_{\beta}^* = I_m \sin(\theta) \end{cases} \quad (23)$$

where the stator current vectors should be circular and have the same amplitude with the case of normal condition. With the limitation of (22), the reference currents of z_1 - z_2 subspace can be calculated by (1), (21), and (23) as

$$\begin{cases} i_{z1}^* = -i_{\alpha} \\ i_{z2}^* = -i_{\beta} \end{cases} \quad (24)$$

It is obvious that the final reference phase currents can be modified by the injection currents in the z_1 - z_2 subspace. Thus, the stable torque is obtained and the amplitude of phase currents is equal during postfault operation. Using the same principles, it is easy to be equivalent when the open-circuit fault occurs in other phases.

The block diagram of the fault-tolerant DTC scheme is shown in Fig. 5. It can be seen that the same controller structure can still be used under the fault-tolerant control, with only minor modifications of current reference value. Meanwhile, the four-vector space vector modulation for the dual three-phase PMSM is adopted. Due to the influence of nonlinear factors such as inverter dead time effect, two proportional integral (PI) controller are applied separately to minimize the harmonic current in z_1 - z_2 subspace. In the fault-tolerant DTC scheme, the key input are the variation of stator flux and torque. By using the electrical angular position, the transformation from u_d , u_q to u_{α} , u_{β} can be expressed as

$$\begin{bmatrix} u_{\alpha} \\ u_{\beta} \end{bmatrix} = \begin{bmatrix} \cos \hat{\theta}_e & -\sin \hat{\theta}_e \\ \sin \hat{\theta}_e & \cos \hat{\theta}_e \end{bmatrix} \begin{bmatrix} u_d \\ u_q \end{bmatrix} \quad (25)$$

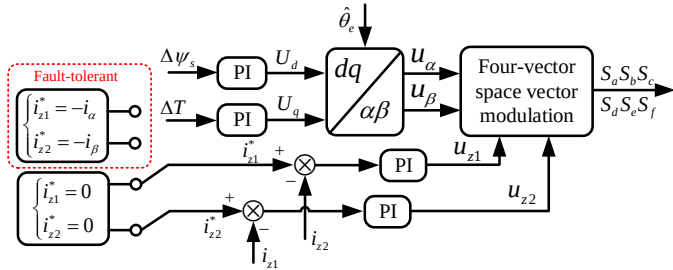


Fig. 5. Structure block diagram of fault-tolerant DTC.

B. Position Estimation

According to (5), the position information can be extracted from the rotor fluxes. Since the electrical angular frequency is the derivative of electrical angular position, the noise of the position signal will be amplified. In order to improve estimation performance, the normalized phase-locked loop (PLL) is adopted. The normalized PLL can reduce the influence of input signal amplitude variation, which is suitable to estimate the smooth electrical angular position.

Fig. 6 shows the structure block diagram of normalized PLL. The input error can be calculated as

$$\varepsilon = \psi_{pm} \sin(\theta_e - \hat{\theta}_e) \approx \psi_{pm} \Delta\theta \quad (26)$$

where $\Delta\theta = \theta_e - \hat{\theta}_e$, $\hat{\theta}_e$ represents the estimated position.

It is worth noting that (26) can be linearized when the position error between the real position and estimated position is insignificant. By analyzing the structure of normalized PLL, the transfer function between the estimation position and the real position can be expressed as

$$G_{PLL} = \frac{\hat{\theta}_e}{\theta_e} = \frac{k_p s + k_i}{s^2 + k_p s + k_i} \quad (27)$$

where k_p and k_i are the parameters of the PI controller in the PLL. It can be concluded that the normalized PLL will have sufficiently good performance with appropriate parameters.

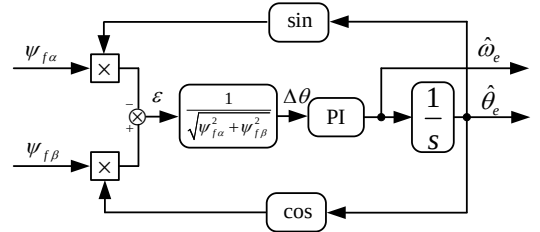


Fig. 6. Structure diagram of normalized PLL.

V. EVALUATION AND VERIFICATION

A. Simulation Results

To verify the proposed sensorless DTC system, a 10 kW dual three-phase PMSM model is built and the simulation results are presented. The proposed sensorless DTC system is shown in Fig. 7. Two cases are investigated in the simulation. In the first case, the stator flux and rotor flux estimated performance of the proposed flux observer are studied. Meanwhile, the estimated stator flux results by using (10) and the rotor flux results by using (11) are investigated for comparison.

Fig. 8 shows the comparison of flux estimation based on 0.5 A current dc bias. It should be noted that 0.5 A bias is superimposed on the currents of $\alpha\beta$ -axis. As shown in Fig. 8(a), the stator flux of $\alpha\beta$ -axis decreases continuously due to the current dc bias. However, the decreasing stator flux will cause the failure of flux synthesis and torque calculation, which brings instability to the DTC system. The stator flux performance of the proposed flux observer with current dc bias is presented in Fig. 8(b). It can be seen that the stator fluxes of $\alpha\beta$ -axis are very sinusoidal and maintains a stable magnitude. The rotor fluxes in Fig. 8(c) show the same trend with Fig. 8(a). At time 0.3 s, the minimum amplitude of rotor fluxes are less than -1 Wb. As shown in Fig. 8(d), the rotor fluxes based on the proposed flux observer have no slope disturbance. The maximum amplitude of rotor fluxes are stable at 0.88 Wb during the steady state. Fig. 9 shows the comparison of flux estimation based on 10 V voltage dc bias. It should be illustrated that the 10 V voltage dc bias is additionally added on the voltages of α -axis and β -axis to verify the performance of flux observer. As shown in Figs. 9(a) and 9(c), the estimated stator flux and rotor flux of α -axis is seriously affected by the

voltage dc bias. Under the action of integrator, the estimated stator flux and rotor flux increases rapidly, which has greatly exceeded the rated value of flux linkage. Meanwhile, the estimated stator fluxes and rotor fluxes based on the proposed flux observer under the same condition are presented in Figs. 9(b) and 9(d). It can be seen that the waveform are stable and without dc bias in both the estimated stator fluxes and rotor fluxes results.

Based on the position observer with the rotor fluxes, the position estimation performance is studied in the second case. Fig. 10 depicts that the simulated results of the speed and position. The real speed and the estimated speed under speed step condition is compared in Fig. 10(a). It can be seen that the estimated speed matches well with the real one at the steady state. When the given speed step occurs from 100 r/min to 200 r/min at 0.5 s, the estimated speed follows the real speed quickly and becomes stable at 200 r/min after a small overshoot. Fig. 10(b) shows the speed error only fluctuates during the speed switching and maintains in a small value in the steady

state. As shown in Fig. 10(c), the estimated position has satisfactory tracking performance with the real position. Additionally, the position estimation performance can be further confirmed by the position error, as shown in Fig. 10(d). Fig. 11 shows the simulation results under healthy, open-circuit fault and fault-tolerant conditions. The given speed is set as 100 r/min, and the load torque is 100 N·m. The phase F is disconnected under the fault condition. In Fig. 11(a), the phase currents has large distortion when the motor is switched from healthy status to fault status. Meanwhile, the torque ripple has exceed 50%. It can be seen that the current waveforms becomes sinusoidal under the fault-tolerant control. The torque ripple is greatly reduced, and stabilize at 100 N·m eventually. As shown in Fig. 11(b), there is no distortion in the waveform of stator fluxes and rotor fluxes under the healthy and fault-tolerant condition. It can be seen that the estimated position matches well with the real position, as shown in Fig. 11(c). The motor speed oscillated during the fault condition, and gradually stabilizes when the motor is switched to fault-tolerant control.

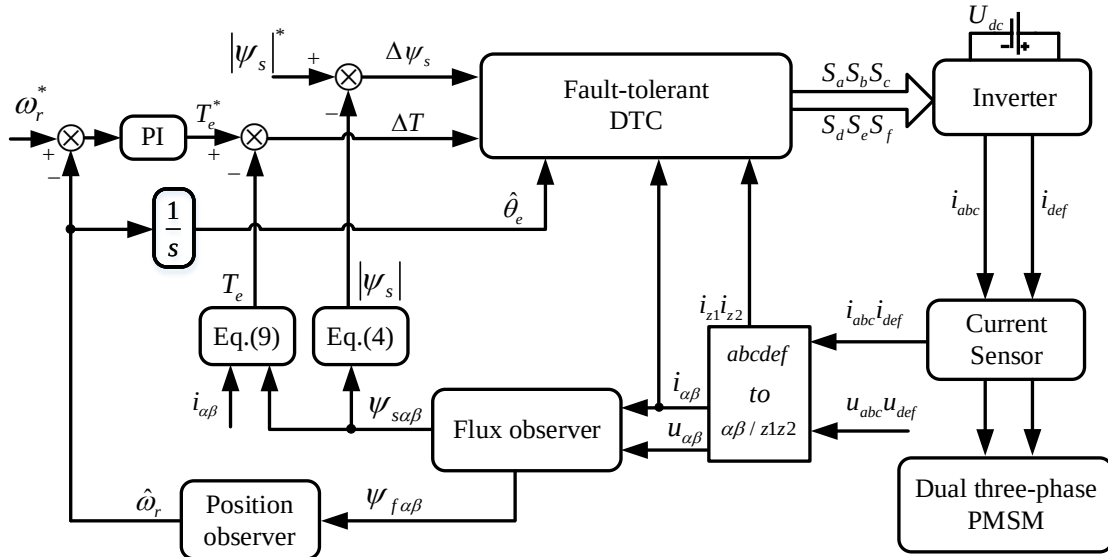


Fig. 7. Block diagram of proposed sensorless DTC scheme for dual three-phase PMSM.

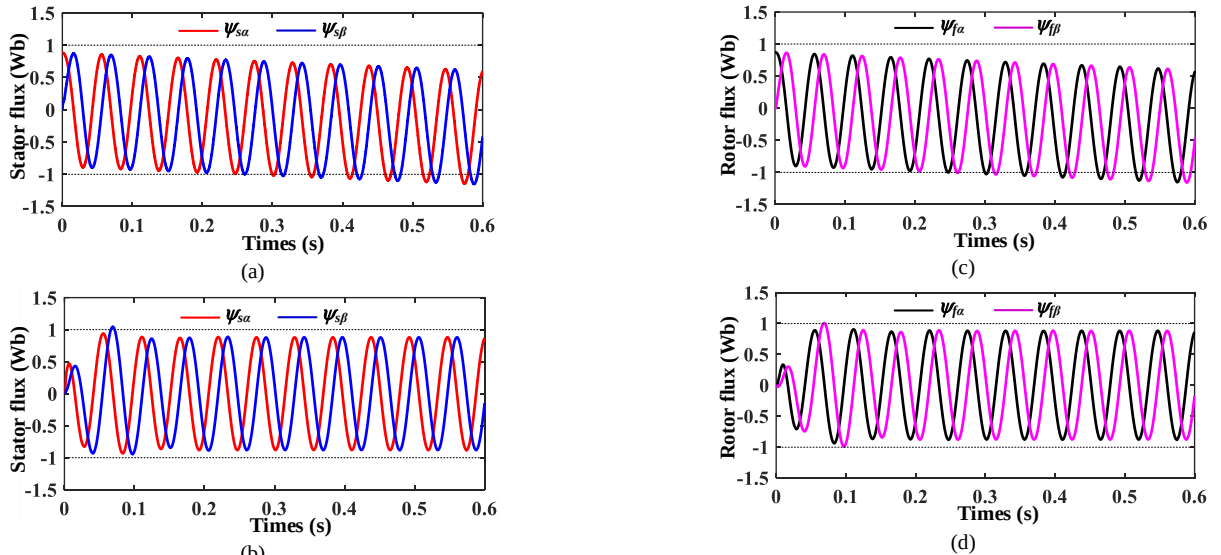


Fig. 8. Comparison of flux estimation based on 0.5A current dc bias. (a) Stator flux by using (10). (b) Stator flux with proposed observer. (c) Rotor flux by using (11). (d) Rotor flux with proposed observer.

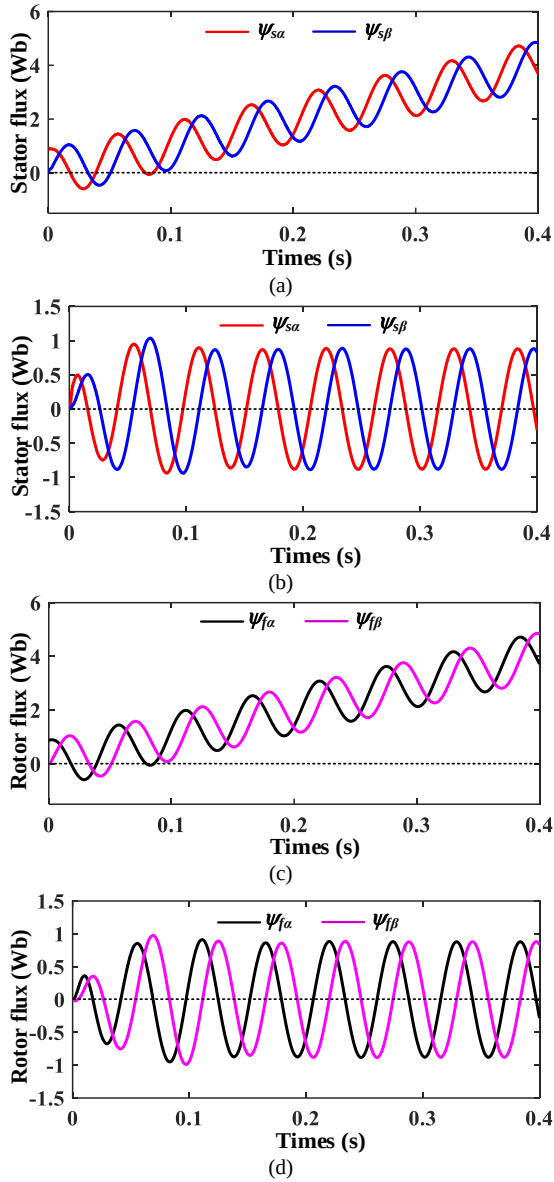


Fig. 9. Comparison of flux estimation based on 10V voltage dc bias. (a) Stator flux by using (10). (b) Stator flux with proposed observer. (c) Rotor flux by using (11). (d) Rotor flux with proposed observer.

Parameters	Value
Motor power	10.8 kW
Pole pair number	11
Stator resistance	0.92 Ω
d-axis inductance	15.21 mH
q-axis inductance	15.74 mH
Permanent flux linkage	0.88 Wb
Rated speed	200 r/min
Rated current	11 A
Rated frequency	59.4 Hz
Rated torque	320 N·m
DC-link voltage	400V

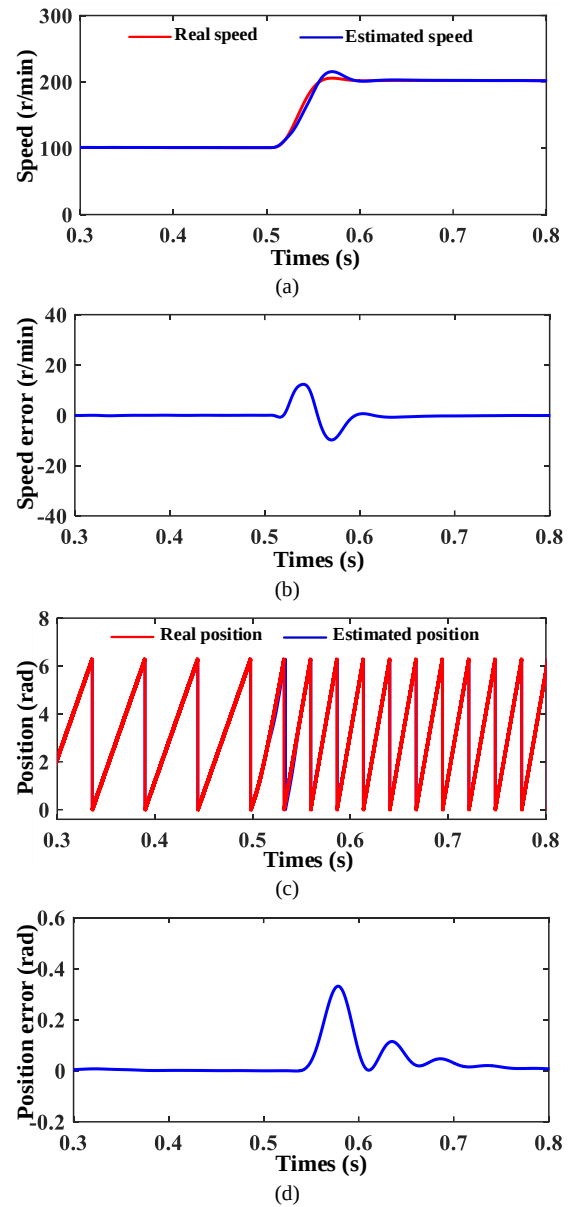
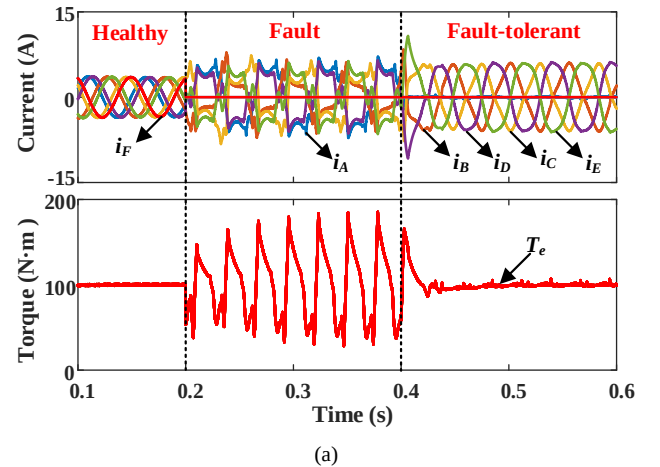


Fig. 10. Simulation results of speed and position. (a) Real speed and estimated speed. (b) Speed error. (c) Real position and estimated position. (d) Position error.



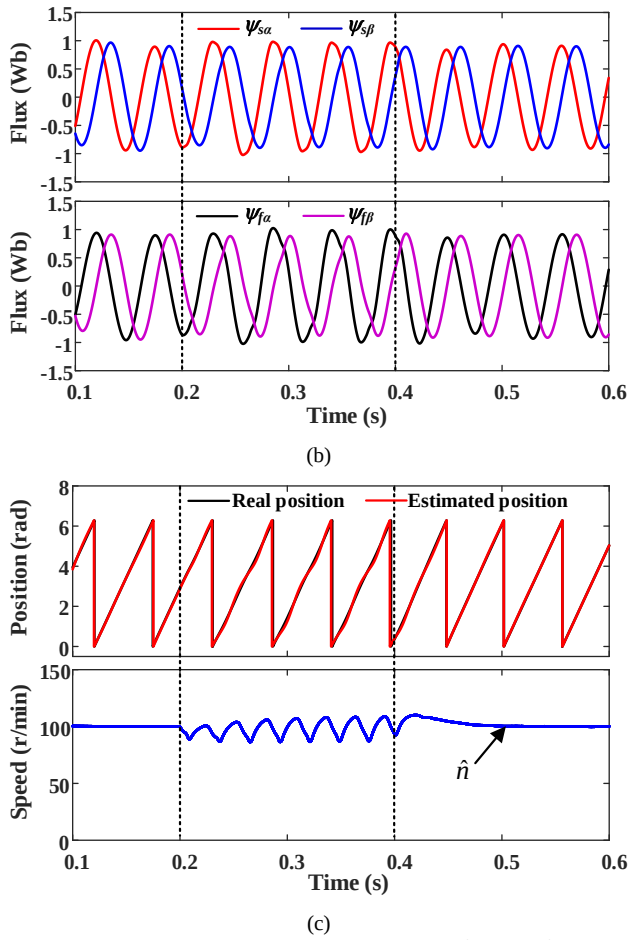


Fig. 11. Simulation results under healthy, open-circuit fault and fault-tolerant conditions. (a) Current and torque. (b) Stator fluxes and rotor fluxes. (c) Real position, estimated position, and speed.

B. Experimental Verification

An experimental platform is established to verify the performance of the sensorless DTC scheme based on the proposed flux observer, as shown in Fig. 12. A DSP TMS320F28377 is used to implement the algorithm. A magnetic powder brake is mechanically coupled with a dual three-phase PMSM as the load. The mechanical speed and rotor position are measured by using a reluctance resolver. It should be noted that the real speed and position obtained from the encoder are only used for comparison. The sampling frequency and pulse width modulation switching frequency are set to 10 kHz. The output voltage of DC source is set as 400 V. The integral gain and proportional gain of the PI controllers in the flux, torque, current in z_1 -axis, and current in z_2 -axis are set as 10 and 800, 2 and 0.05, 50 and 150, 50 and 150, respectively. In the flux observer, the feedback gain is -0.92. The parameters of PLL are set as $k_p = 500$ and $k_i = 5$.

Fig. 13 shows the experimental results based on the proposed flux observer. The motor operating at 100 r/min under no load condition. As shown in Fig. 13(a), it can be realized that the estimated stator fluxes have satisfactory performance due to the sinusoidal waveform. The stator flux circle is synthesized by stator fluxes of $\alpha\beta$ -axis, which has no distortion and maintains a stable value. The estimated rotor fluxes of $\alpha\beta$ -axis are shown in

Fig. 13(b). Similarly, the rotor fluxes have no distortion and the maximum amplitude is stable at 0.88 Wb.

The stator flux and rotor flux estimation performance with the disturbances of dc bias are involved. Fig. 14 shows the experimental results when the voltage and current dc bias are applied. The stator flux and rotor flux of α -axis are chosen to compare the performance with and without dc bias. In Fig. 14(a), an additional 40 V voltage dc bias is superimposed on the voltages of α -axis and β -axis, which is just for comparison. It can be seen that the waveforms of stator flux and rotor flux have no difference when the voltage dc bias is applied. Compared with the zoom1 and zoom2 in Fig. 14(a), the amplitude of stator flux and rotor flux maintains in a stable value with or without the voltage dc bias. Meanwhile, 0.5 A current dc bias is superimposed on the currents of α -axis and β -axis. As shown in Fig. 14(b), both the stator flux and rotor flux have similar performance when the current dc bias is applied. It can be seen that the stator flux and rotor flux are sinusoidal waveforms. The waveforms have no distortion when 0.5 A current dc bias is added. As a result, the proposed flux observer has the ability to eliminate the dc bias whether it is voltage bias or current bias.

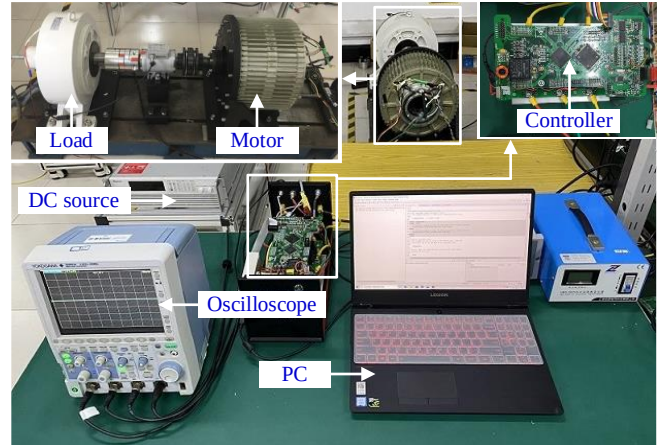
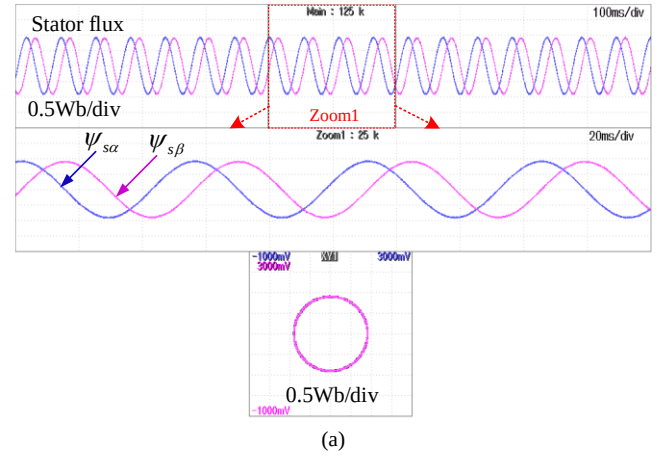


Fig. 12. Experimental platform.



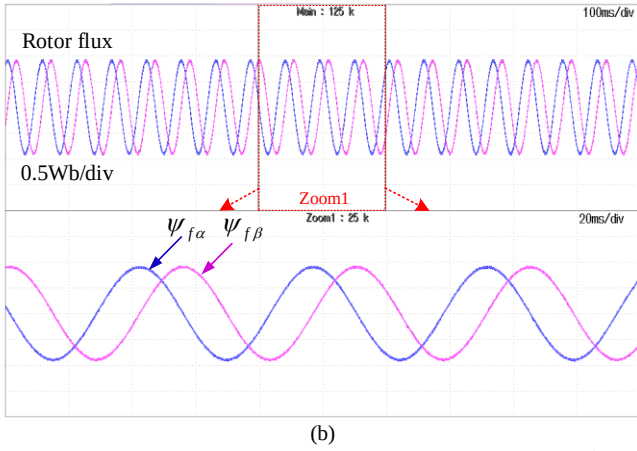


Fig. 13. Experimental results based on proposed observer. (a) Stator flux. (b) Rotor flux.

The position estimation performance in the steady state and dynamic process are also verified by the experimental results. Fig. 15 shows the experimental result of the speed step under 50% rated load. The reference speed changes from 100 r/min to 200 r/min which is corresponding to the half rated speed and the rated speed, respectively. The waveforms of the real speed, estimated speed, and speed error are depicted in Fig. 15(a). It can be seen that the estimated speed can track the real speed very quickly. The speed error only fluctuates in the speed step and keeps in a small value in the steady state. The waveform of real position, estimated position, and position error are shown in Fig. 15(b). It can be seen that the estimated position matches well with the real position and the position estimation error is less than 0.08 rad. These results verify that the proposed sensorless DTC scheme has a good performance during the steady state and the speed step process.

Fig. 16 shows the experimental results under variable load torque condition. The motor is operating at 100 r/min, and the load torque is switching from the rated load to the no load condition, then to the rated load condition. In Fig. 16(a), the torque and speed of motor are stable at the rated condition which is 320 N·m and 200 r/min. When the load torque changes from the rated load to no load condition, the current (phase A) is also decreasing accordingly. The amplitude of phase current is decreased from 11 A to 1 A. The motor speed fluctuated only when the torque changed, and finally stabilized at 100 r/min. In Fig. 16(b), the position estimation performance under variable load condition is presented. The estimated position is closed with the real position, and remains relatively stable in steady state. The position error fluctuates briefly when the load changes. Due to the influence of nonlinear factors such as parameter saturation under heavy load, the position error under the rated load condition will be larger than that under the no load condition. The position error is stable at 0.2 rad and 0.1 rad under the rated load and no load condition, respectively.

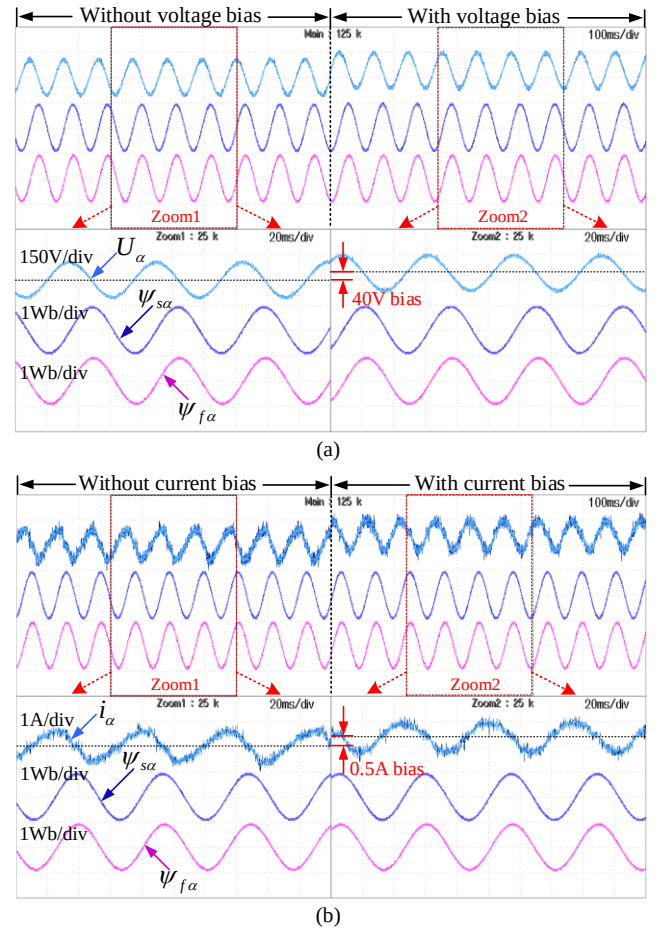


Fig. 14. Experimental results of dc bias. (a) With and without 40V voltage bias. (b) With and without 0.5 A current bias.

Fig. 17 shows the experimental results under healthy, open-circuit fault and fault-tolerant conditions. The motor is operating at 100 r/min with 100 N·m load torque. It should be noted that the phase F is disconnected when open-circuit fault occurs. As shown in Fig. 17(a), the torque ripple becomes very large under the fault condition. The average torque ripple has exceeded 40%. The current amplitude of phase B increases and distortion occurs at the same time. When fault-tolerant control scheme is applied, the torque ripple decreases rapidly. Meanwhile, the harmonic content in the phase current is reduced, and the amplitude of phase current is stable at 6 A. It can be seen that the waveform of stator flux and rotor flux hardly fluctuates even in the switching process. In Fig. 17(b), the current of phase F step from the maximum value 3.5 A to 0 A when the open-circuit fault occurs. The estimated position can track the real position well. The motor speed fluctuated slightly during switching, and finally stabilized at 100 r/min. The online switching experiments indicate that the proposed method is feasible, and the performance of position estimation and torque control is well presented.

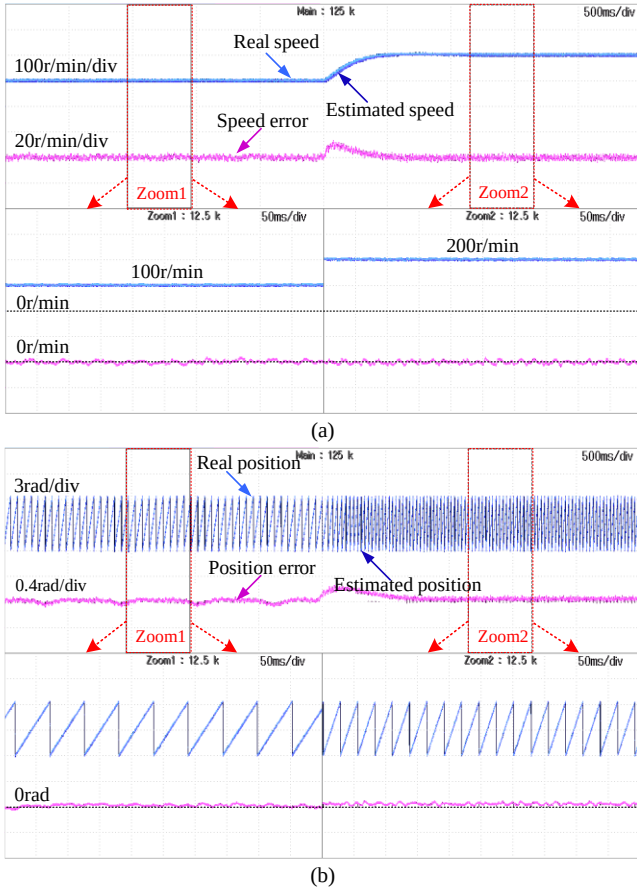


Fig. 15. Experimental results of speed step. (a) Real speed, estimated speed, and speed error. (b) Real position, estimated position, and position error.

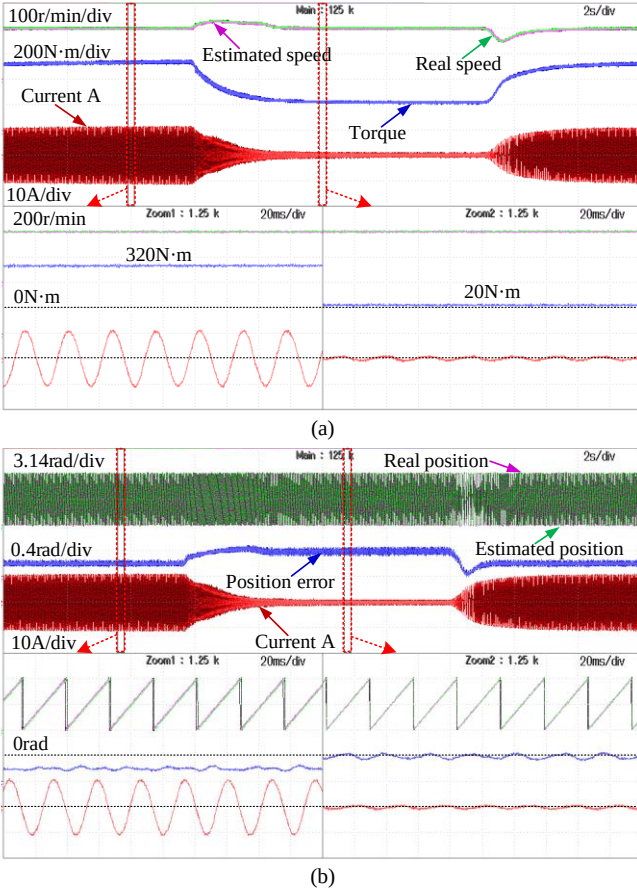


Fig. 16. Experimental results under variable load torque condition. (a) Real Speed, estimated speed, torque, and current. (b) Real position, estimated position, position error, and current.

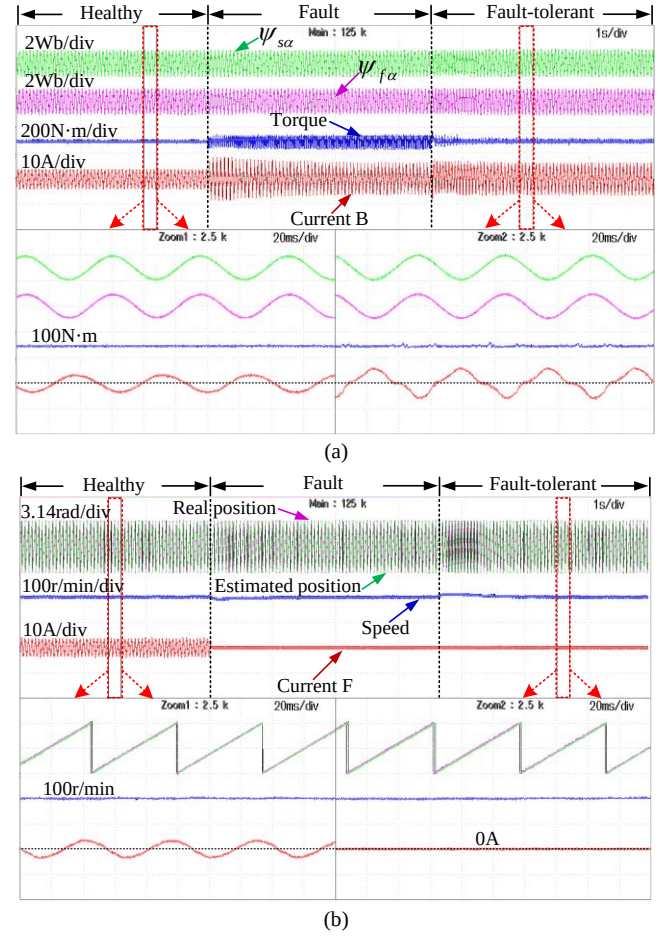


Fig. 17. Experimental results under healthy, open-circuit fault and fault-tolerant condition. (a) Stator flux, rotor flux, torque, and current (phase B). (b) Real position, estimated position, speed, and current (phase F).

VI. CONCLUSION

This paper has proposed a modified flux observer for sensorless DTC scheme of the dual three-phase PMSM. The proposed flux observer has no phase lag and no attenuation, which is more accurate and robust than the conventional flux calculation method. The disturbance of dc bias and high frequency noise caused by the measurement and parameter fluctuation can be significantly reduced. By utilizing the position observer, the rotor position information for sensorless operation can be achieved. A fault-tolerant control scheme based on normal decoupling transformation matrix is introduced. The performance of torque control and position estimation are evaluated under the healthy, open-circuit fault, and fault-tolerant conditions. Simulation and experiment have been conducted to verify the proposed method. The simulation and experimental results show that the estimated performance of stator flux and rotor flux can be improved by the proposed flux observer. It is also verified that the proposed sensorless DTC system for dual three-phase PMSM has the effectiveness and good performance even under the open-circuit fault condition.

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